Malaria Patient Analytics Dashboard

Comprehensive Analysis of Malaria Cases and Predictive Insights

Patients Overview

Total Patients

1622

Average Age of Patients

in years

44.71

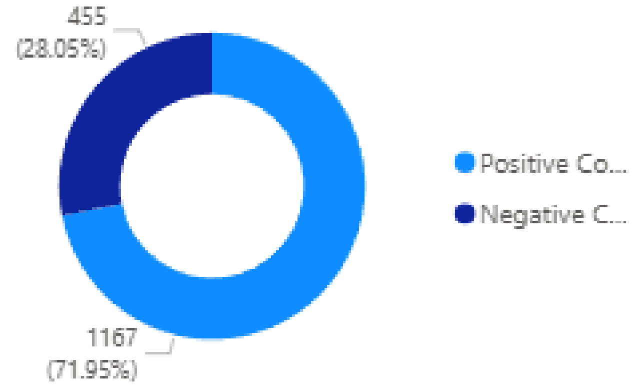
Patients Tested Positive Count

1167

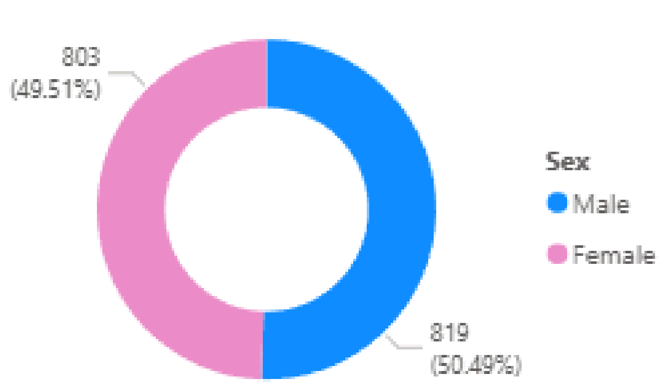
Patients Tested Negative Count

455

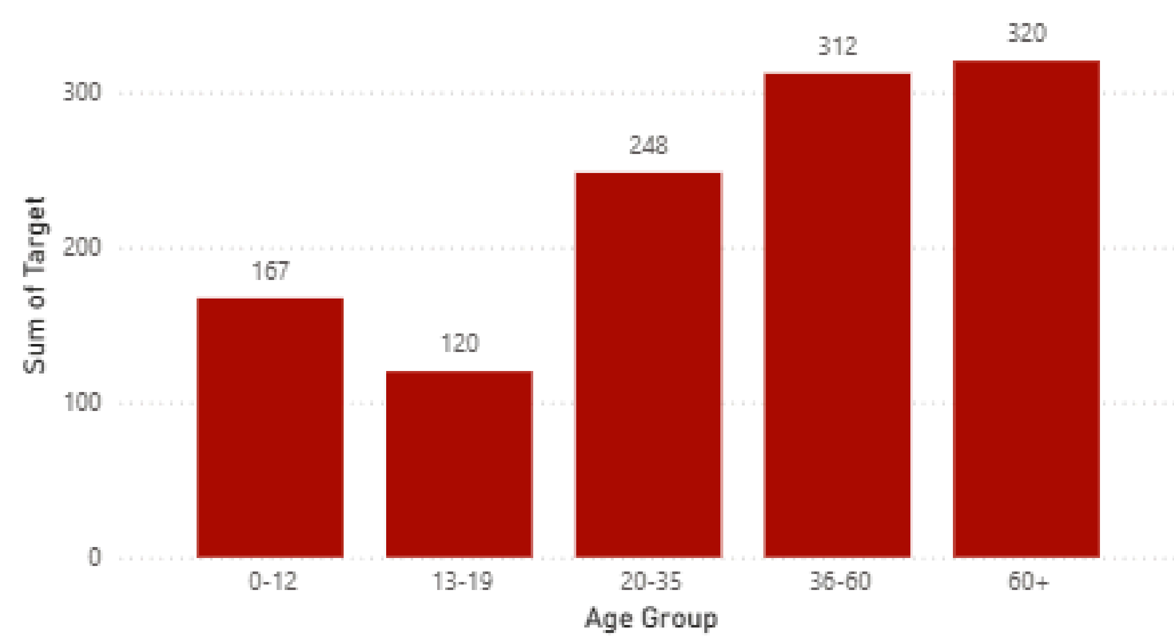
Positive and Negative Cases of Malaria Ratio



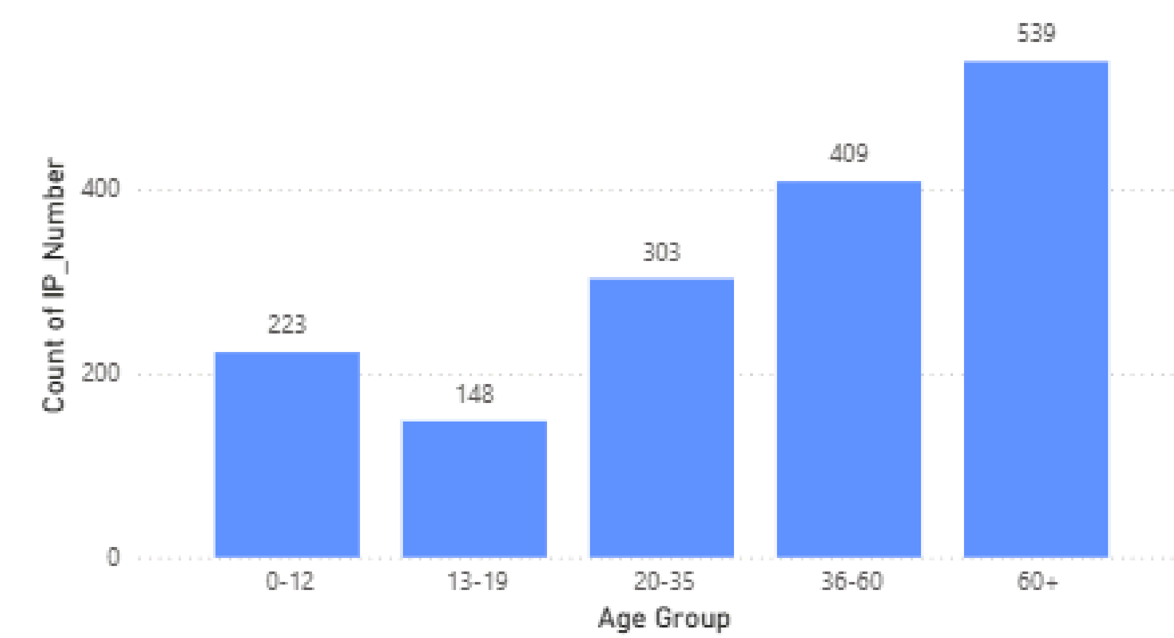
Ratio of Patients based on Sex



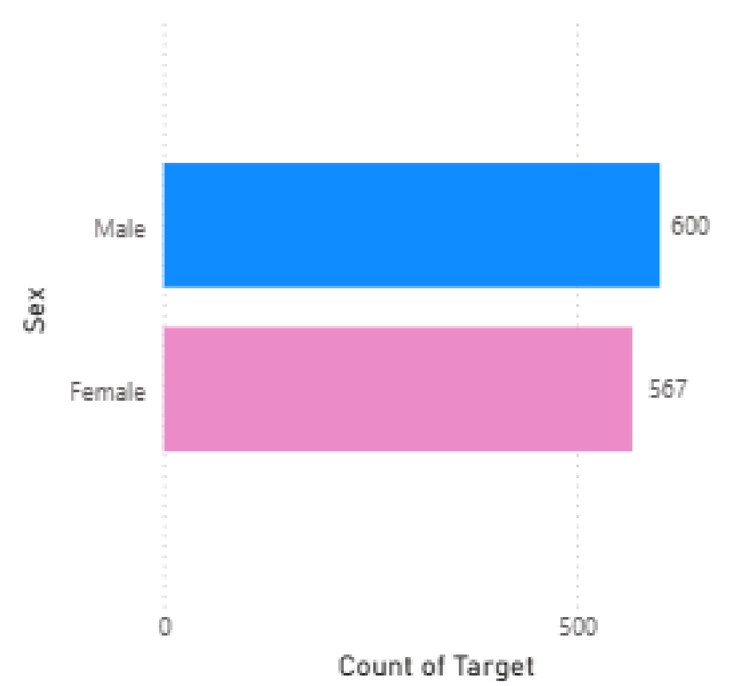
Total Positive Cases by Age Group



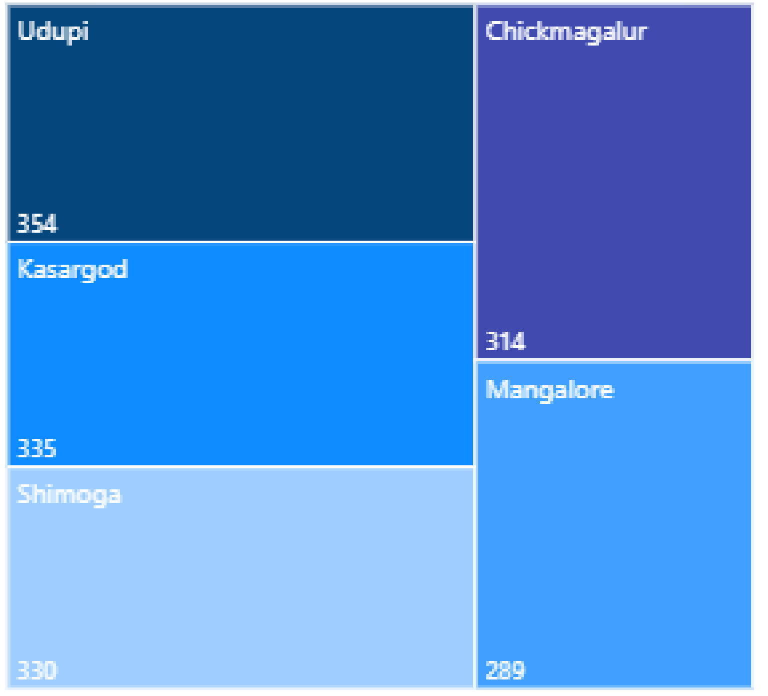
Total Patients by Age Group



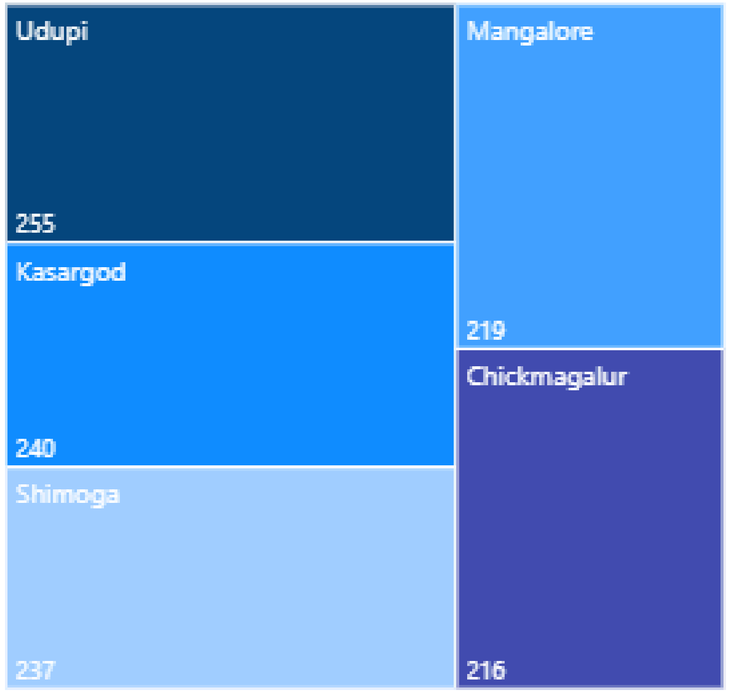
Positive Cases by Sex



Patients per Residence Area



Positive Cases per Residence Area



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Comprehensive Analysis of Malaria Cases and Predictive Insights

Diagnostic Insights & Model Accuracy

Number of Identifiable Symptoms

11

Count of Attribute

Average Symptoms Count per Patient

5.45

Average of Symptom Count

Model Accuracy in Predicting Malaria

94.61

Model Accuracy (%)

Number of Patients used as Training Data

973

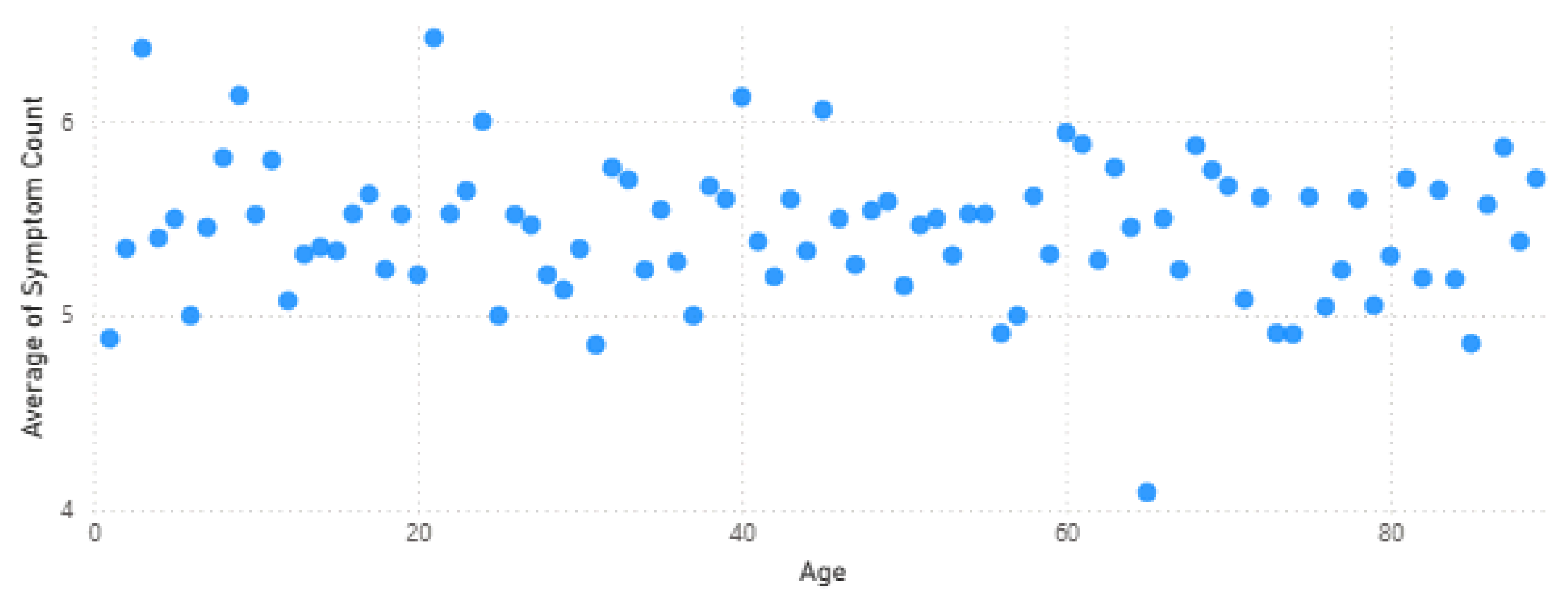
Testing Data

Number of Patients Tested using Prediction Model

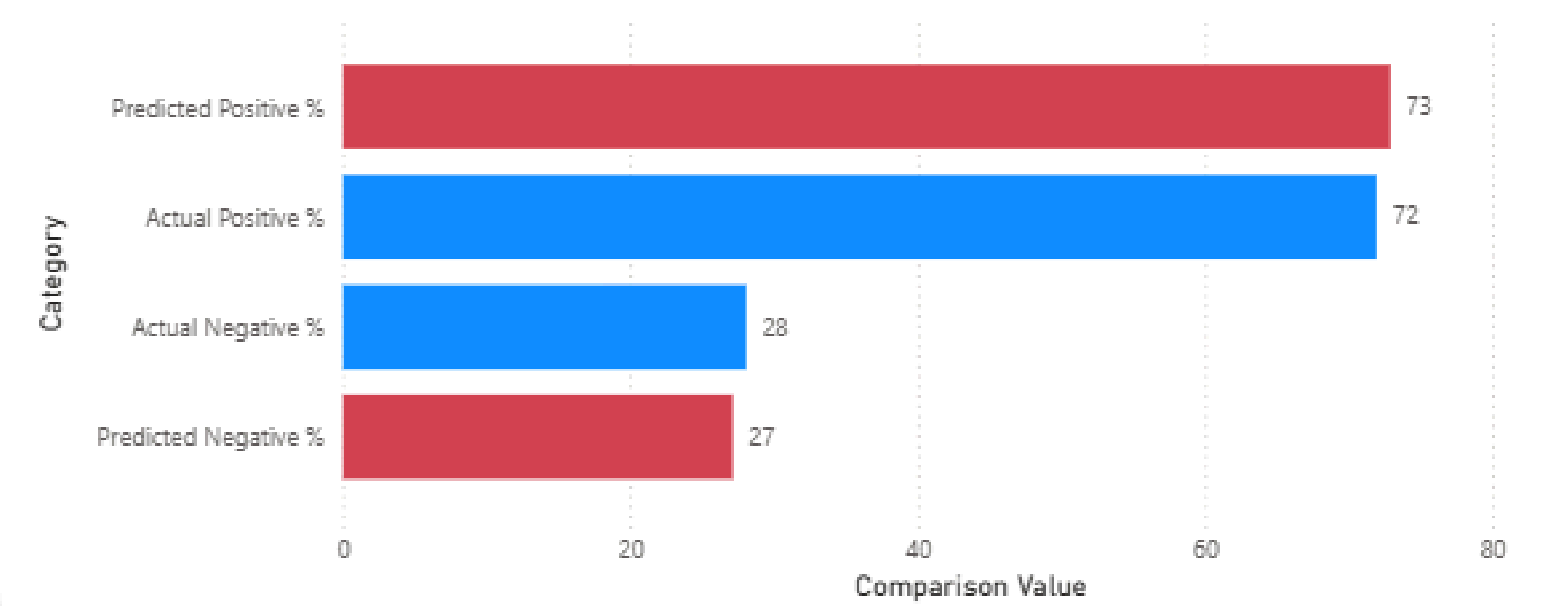
649

Count of Row No.

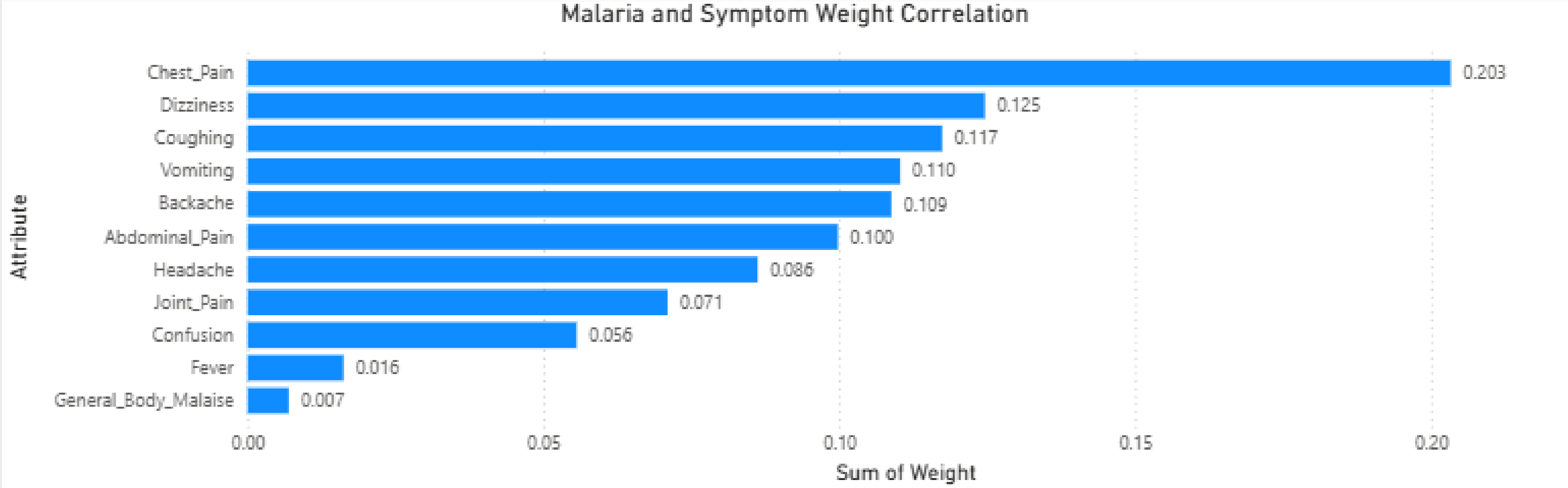
Age and Symptoms Count Correlation of Tested Positive Patients



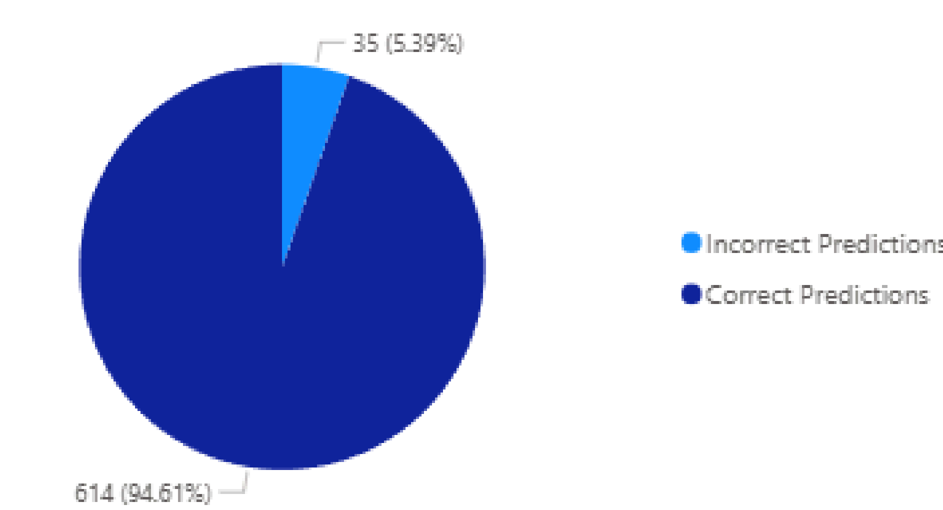
Comparison of Actual vs Predicted Percentage of Patient Diagnosis



Malaria and Symptom Weight Correlation



Correct and Incorrect Predictions Ratio based on Model Confidence Level



Objective 1: To assess the overall number of malaria cases and the balance between positive and negative diagnoses in the dataset.

Objective 2: To analyze how malaria affects different demographic and geographic groups by examining patient age, gender, and residential area distribution.

Objective 3: To identify the overall symptom profile of malaria by analyzing the number of identifiable symptoms and the average symptoms per patient.

Objective 4: To determine which symptoms contribute the most to predicting malaria based on their weight or importance in the model.

Objective 5: To evaluate the performance and reliability of the malaria prediction model through accuracy testing and comparison of predicted versus actual diagnosis outcomes.

Objective 6: To examine how patient age relates to the number of malaria symptoms experienced to identify any age-based differences in symptom presentation.

Objective 1:
To assess the overall number of malaria cases and the balance between positive and negative diagnoses in the dataset.
KPI Requirement:

- **KPI:** Total Patients, Patients Tested Positive Count, Patients Tested Negative Count, and Positive and Negative Cases of Malaria Ratio
- **Description:** Represents total patient records and the proportion of positive versus negative malaria results.
- **SMART Requirement:**
    - **Specific:** Count all total, positive, and negative patient entries in the dataset.
    - **Measurable:** Total = 1,622; Positive = 1,167; Negative = 455; Ratio = 71.95% positive, 28.05% negative.
    - **Achievable:** Derived from the final dataset used for malaria diagnosis modeling.
    - **Relevant:** Reflects overall malaria prevalence within the studied population.
    - **Time-Bound:** Evaluated per data collection cycle.

Objective 2:
To analyze how malaria affects different demographic and geographic groups by examining patient age, gender, and residential area distribution.
KPI Requirement:

- **KPI:** Average Age of Patients, Ratio of Patients based on Sex, Positive Cases by Sex, Total Positive Cases by Age Group, Total Patients by Age Group, Patients per Residence Area and Positive Cases per Residence Area
- **Description:** Displays the demographic and spatial patterns of malaria cases across different ages, sexes, and locations.
- **SMART Requirement:**
    - **Specific:** Group malaria cases based on age, gender, and area to identify trends.
    - **Measurable:** Calculated using counts and percentages across categories.
    - **Achievable:** Derived from demographic and geographic data in the dataset.
    - **Relevant:** Supports targeted malaria awareness and prevention strategies.
    - **Time-Bound:** Evaluated once per reporting period or dataset update.

Objective 3:
To identify the overall symptom profile of malaria by analyzing the number of identifiable symptoms and the average symptoms per patient.
KPI Requirement:

- **KPI:** Number of Identifiable Symptoms and Average Symptoms Count per Patient
- **Description:** Shows the total distinct symptoms in the dataset and the mean number of symptoms reported per patient.
- **SMART Requirement:**
    - **Specific:** Identify all recorded symptoms and calculate the average number per patient.
    - **Measurable:** Total identifiable symptoms and mean symptom count per Patient (average ≈ 5.45).
    - **Achievable:** Computed from the collected malaria patient dataset.
    - **Relevant:** Helps understand the common symptom patterns seen in malaria cases.
    - **Time-Bound:** Evaluated during the data analysis phase of the study.

Objective 4:
To determine which symptoms contribute the most to predicting malaria based on their weight or importance in the model.
KPI Requirement:

- **KPI**: Malaria and Symptom Weight Correlation
- **Description:** Shows the contribution level of each symptom to the model's malaria prediction results.
- **SMART Requirement:**
  - **Specific:** Rank symptoms according to their predictive weight in the model.
  - **Measurable:** Weights measured as numerical feature importance values (e.g., Chest Pain = 0.203).
  - **Achievable:** Extracted from AI Studio's model analysis of feature contribution.
  - **Relevant:** Identifies key symptoms that have the highest diagnostic impact.
  - **Time-Bound:** Evaluated after final model training is completed.

Objective 5:
To evaluate the performance and reliability of the malaria prediction model through accuracy testing and comparison of predicted versus actual diagnosis outcomes.
KPI Requirement:

- **KPI:** Model Accuracy in Predicting Malaria, Number of Patients Used As Training Data, Number of Patients Tested Using Prediction Model, and, Correct and Incorrect Predictions Ratio Based on Model Confidence Level. Comparison of Actual vs Predicted Percentage of Patient Diagnosis.
- **Description:** Measures how accurately the model detects malaria, including data used for training and testing, prediction correctness, and how close predicted results are to real diagnoses.
- **SMART Requirement:**
    - **Specific:** Assess the model’s accuracy and verify how closely predictions match real outcomes.
    - **Measurable:** Accuracy = 94.61%; Training = 973; Testing = 649; Predicted Positive = 73%; Actual Positive = 72%;
    - **Achievable:** Metrics taken from AI Studio’s model training.
    - **Relevant:** Confirms the model’s dependability in predicting malaria accurately.
    - **Time-Bound:** Evaluated after every training and testing phase.

Objective 6:
To examine how patient age relates to the number of malaria symptoms experienced to identify any age-based differences in symptom presentation.
KPI Requirement:
- **KPI:** Age and Symptoms Count Correlation of Tested Positive Patients
- **Description:** Displays the relationship between a patient's age and the number of malaria symptoms they exhibit.
- **SMART Requirement:**
  - **Specific:** Analyze how the number of symptoms changes across different age ranges.
  - **Measurable:** Correlation values between age and symptom count.
  - **Achievable:** Computed from patient age and symptom data in the dataset.
  - **Relevant:** Helps identify whether certain age groups experience more or fewer symptoms.
  - **Time-Bound:** Analyzed once per dataset update or model retraining.

SUMMARY:

The report aims to analyze malaria cases using data-driven modeling to understand patterns, risk factors, and predictive symptoms. It assessed the overall number of malaria cases over 1622 patients, showing that 71.95% (1,167) of patients tested positive and 28.05% (455) tested negative. The demographic analysis examined how factors like age, gender, and location affected malaria prevalence, showing that males had slightly more positive malaria cases than females.

Symptom analysis revealed that patients reported an average of 5.45 symptoms, helping to identify common malaria indicators. The model determined which symptoms most strongly predicted malaria, ranking them based on their feature weights in AI studio and revealing chest pain as the most influential factor, contributing about 20% to the likelihood of malaria. Model performance testing achieved a 94.61% accuracy, confirming the reliability of the malaria prediction model. The Actual vs. Predicted Comparison chart showed the alignment between predicted positive (73%) and actual positive (72%) cases, as well as predicted negative (27%) and actual negative (28%) cases, confirming high predictive reliability. Finally, the study found a correlation between age and symptom count, the data suggests that malaria symptoms are fairly consistent across all ages, showing that age is not a significant factor in patients who are tested positive.

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Source of Dataset Used

DATASET:

https://www.kaggle.com/datasets/programmer3/malaria-diagnosis-dataset/data

Malaria Diagnosis Dataset

Symptoms & Demographics

Data Card Code (0) Discussion (0) Suggestions (0)

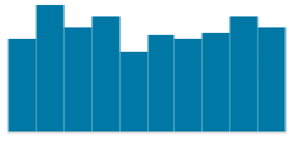
About Dataset

This dataset contains 1,622 patient records. It integrates demographic features (age, sex, residence area, admission/discharge dates) and clinical symptoms (fever, headache, abdominal pain, general body malaise, dizziness, vomiting, confusion, backache, chest pain, coughing, and joint pain).

The dataset includes demographic features such as Age, Sex, and Residence Area, which help assess malaria risk in different population groups. Admission (DOA) and Discharge Dates provide temporal context for patient care. Clinical symptoms include Fever, Headache, Abdominal Pain, General Body Malaise, Dizziness, Vomiting, Confusion, Backache, Chest Pain, Coughing, and Joint Pain, which are commonly reported in malaria cases and vary in severity. Primary Code and Diagnosis Type specify the malaria classification. The Target column indicates malaria presence (1) or absence (0).

Malaria_Dataset.csv (208.11 kB)

Detail Compact Column 10 of 21 columns

IP_Number	Age	Sex		Residence_Area		DOA	Discha
1473 unique values		Male	50%	Udupi	22%	1622 unique values	unic
		Female	50%	Kasargod	21%		
				Other (933)	58%		

Usability 8.76

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Expected update frequency Not specified

Tags Social Science

Data Explorer Version 1 (208.11 kB)

Malaria_Dataset.csv

Summary

1 file

21 columns